OpenClaw Model Selection — 48GB VRAM Tier

Part 5: Native 48GB Card & 3× RTX 5060 Ti 16GB Pipeline Split

> **Critical constraint:** Every model listed as viable must load weights AND KV cache 100% in VRAM. No CPU offloading, no partial offloading, no hybrid RAM modes. These configurations must work even on PCIe 2.0 where any offloading would kill usability.

Executive Summary

Forty-eight gigabytes of VRAM crosses a critical threshold: the 70-billion-parameter class. Llama 3.3 70B, DeepSeek-R1-Distill-70B, and Llama-Nemotron-Super-49B all load fully within this budget, bringing BFCL tool-calling scores of 77%, AIME 2024 reasoning at 70%, and HumanEval code generation at 88% — numbers no 32B model reaches. Every 27–36B model from the 32GB tier now runs at near-lossless Q8_0 quantization with expanded context windows.

Does 48GB unlock genuinely new capabilities for OpenClaw? **Yes — unambiguously.** But with a sharp caveat: the 3×16GB pipeline-split path delivers 70B models at a painful 7–10 t/s, making the native single-card path strongly preferable for interactive agent work.

This analysis covers two deployment paths:

- **(a) Native 48GB GPU** (RTX A6000, RTX 6000 Ada)
- **(b) 3× RTX 5060 Ti 16GB** pooling 48GB via pipeline parallelism

Both paths access the same model roster but at dramatically different speeds.

---

1. New Models That Unlock at 48GB

The jump from 32GB to 48GB adds a small but transformative set of models. Only models that fit weights + usable KV cache entirely in VRAM are listed.

1.1 Complete 48GB Model Roster

Model	Params	Architecture	Quant	Weight Size	KV Room	Usable Context	Fits 48GB?
Llama 3.3 70B	70B	Dense	Q4_K_M	42.5 GB	5.5 GB	8–16K	✅ Yes
DeepSeek-R1-Distill-Llama-70B	70B	Dense	Q4_K_M	42.5 GB	5.5 GB	8–16K	✅ Yes

Model	Params	Architecture	Quant	Weight Size	KV Room	Usable Context	Fits 48GB?
Nemotron-Super-49B v1.5	49B	Dense (pruned)	Q4_K_M	30.2 GB	17.8 GB	32–64K	✅ Yes
Nemotron-Super-49B v1.5	49B	Dense (pruned)	Q6_K	40.9 GB	7.1 GB	8–16K	✅ Yes

Minimum quantization: Q4_K_M. No sub-Q4 quantizations are considered viable — quality degradation at Q3 and below is unacceptable for production agent work. Q5_K_M for 70B models (49.95 GB) does NOT fit. The only usable quant for true 70B models at 48GB is Q4_K_M.

1.2 Model Details

Llama 3.3 70B Instruct — the headline unlock

Meta's 70B dense model is the single most important addition. At Q4_K_M (42.5 GB), it loads with ~5.5 GB remaining for KV cache — enough for 8–16K context with Q8 KV quantization. Context is tight at this tier, but 8–16K is sufficient for most agentic tasks (tool calls, code generation, orchestration). For roles needing longer context, use Nemotron-Super-49B Q4_K_M (30.2 GB, 32–64K context) instead.

Benchmarks position this well above the 32B class: MMLU 86.0%, HumanEval 88.4%, MATH 77.0%, BFCL v2 77.3% (tool calling), IFEval 92.1%. Native function calling via Llama 3 JSON schema format. The 77.3% BFCL v2 score significantly exceeds any 32B model for agentic tool invocation.

DeepSeek-R1-Distill-Llama-70B — reasoning transformed

Identical Llama 3.3 70B architecture (42.5 GB at Q4_K_M), fine-tuned with reasoning chains from the full DeepSeek-R1 671B. Dramatic reasoning improvement: AIME 2024 70.0% (vs ~50% for 32B distill, ~20% for base Llama 3.3), MATH-500 94.5%, GPQA Diamond 65.2%.

Does NOT support native tool calling. No BFCL scores. This is a pure reasoning engine, not a general-purpose agent model. Pair with a tool-calling model for OpenClaw roles requiring function execution.

Nemotron-Super-49B v1.5 — the unexpected sweet spot

NVIDIA's NAS-optimized derivative of Llama 3.3 70B, pruned to ~49B parameters with skip-attention layers and variable FFN widths. Remarkable VRAM efficiency: Q4_K_M at 30.2 GB leaves 17.8 GB for KV cache (32–64K context), and Q6_K at 40.9 GB still leaves 7.1 GB. Q8_0 at 53 GB does NOT fit.

Outperforms many 70B models on reasoning benchmarks thanks to distillation from larger teacher models. Features reasoning ON/OFF toggle via system prompt. Explicitly supports agentic tasks including RAG and tool calling, with NVIDIA providing a custom vLLM tool-call parser.

For most OpenClaw agent roles, Nemotron-Super-49B at Q4_K_M is the optimal 48GB model — it provides 70B-class quality with room for 32–64K context.

Qwen2.5-72B Instruct — does NOT fit at minimum Q4

Qwen's 72.7B dense model at Q4_K_M reaches 47.4 GB — leaves only 0.6 GB for KV cache, which is effectively unusable. The 152K vocabulary inflates weight size beyond other 70B models. Only viable at Q3_K_M (~39 GB), which violates the minimum Q4 quantization constraint. **Not recommended. Listed in the "doesn't fit" table.**

1.3 Key Benchmarks — 48GB Models vs 32GB Models

Benchmark	Qwen3-32B (32GB tier)	Llama 3.3 70B	R1-Distill-70B	Nemotron-49B	Jump
MMLU	~81%	**86.0%**	~83%	~84%	+3–5%
HumanEval	~78%	**88.4%**	~82%	~85%	+7–10%
MATH	~72%	**77.0%**	**94.5%**	~78%	+5–22%
BFCL v2 (tool calling)	~65%	**77.3%**	N/A	~74%	+9–12%
IFEval	~85%	**92.1%**	~88%	~90%	+5–7%
AIME 2024	~35%	~20%	**70.0%**	~45%	+10–35%
GPQA Diamond	~50%	~53%	**65.2%**	~55%	+5–15%

---

2. 32GB-Tier Models That Improve at 48GB

The 48GB budget transforms existing 27–36B models from "adequate at Q4" to "near-lossless at Q8_0." Quantization quality follows a well-established curve: Q4_K_M adds +0.054 perplexity over FP16, Q6_K adds +0.012, Q8_0 adds +0.001. The jump from Q4 to Q8_0 represents ~1% perplexity improvement that blind evaluations consistently detect at 13B+ scales, particularly in code generation accuracy, reasoning chain stability, and instruction-following precision.

Model	32GB Tier (quant / ctx)	48GB Tier (quant / ctx)	What improves
Qwen3-32B	Q4_K_M (19.8 GB) / ~42K	Q8_0 (34.8 GB) / 48–96K	Near-lossless + 2× context
DS-R1-Distill-Qwen-32B	Q4_K_M (20.1 GB) / ~42K	Q8_0 (34.8 GB) / 48–96K	Critical for reasoning chains
Gemma 3 27B	Q8_0 (28.7 GB) / 4–8K	Q8_0 (28.7 GB) / **full 128K**	19 GB free; 5:1 local-to-global attention = tiny KV
EXAONE 4.0 32B	Q4_K_M (19.3 GB) / ~32K	Q8_0 (34.0 GB) / 32–64K	Quality + context
Seed-OSS-36B	Q4_K_M (21.8 GB) / ~16K	Q6_K (29.7 GB) / 32–64K	Q8_0 tight at 38.4 GB; Q6_K optimal
Qwen3-30B-A3B (MoE)	Q4_K_M (18.6 GB) / ~128K	Q8_0 (32.5 GB) / 160–320K	Near-lossless + massive context
Qwen3-Coder-30B (MoE)	Q4_K_M (18.6 GB) / ~128K	Q8_0 (32.5 GB) / 160–320K	Near-lossless coding quality
Nemotron 3 Nano 30B	Q4_K_M (24.5 GB) / 128K+	Q8_0 (33.6 GB) / 128K+	Quality upgrade; tiny KV needs (6 of 52 layers use attention)
GPT-oss-20B (MoE)	MXFP4 (13.7 GB)	MXFP4 (13.7 GB) + massive ctx room	No quant change needed; more KV headroom

Gemma 3 27B deserves special mention. Its 5:1 local-to-global attention ratio means only ~10 of 62 layers maintain full-context KV cache, reducing per-token cost to ~15–20% of standard transformers. At Q8_0 (28.7 GB) on 48GB, it runs its full **128K native context** comfortably with 19 GB to spare.

---

3. Pipeline Parallelism — How 3-Way Splitting Works

3.1 The Problem: 70B Models Need 3 Cards

At Q4_K_M, Llama 3.3 70B requires 42.5 GB. Two 16GB cards provide 32GB — not enough. Three 16GB cards provide 48GB — the minimum. Pipeline parallelism distributes layers sequentially: GPU 0 processes layers 0–26, GPU 1 handles 27–53, GPU 2 finishes 54–80.

Same principle as 2-way split: only one GPU computes at a time ("pipeline bubble"). Adding a third GPU adds VRAM capacity but does NOT add speed. In fact, the extra pipeline hop slightly reduces throughput vs 2-way.

3.2 MoE vs Dense on 3-Way Split

MoE models don't benefit from 3 GPUs — they already fit on 2 cards (18.6 GB). Adding a third card for an MoE model is wasteful; it adds a pipeline stage that only slows things down.

Dense 70B models REQUIRE 3 cards but suffer the full pipeline penalty: each token must traverse all 3 GPUs sequentially. Expected speed for Llama 3.3 70B Q4_K_M on 3× RTX 5060 Ti: **~7–10 t/s**.

3.3 Per-Model Split Behavior on 3× 16GB

Model	Weight Size	Cards Needed	Speed (3× 5060 Ti)	Speed Retention
Llama 3.3 70B Q4_K_M	42.5 GB	3	~7–10 t/s	~25% (dense, 3-way)
R1-Distill-70B Q4_K_M	42.5 GB	3	~7–10 t/s	~25% (dense, 3-way)
Nemotron-Super-49B Q4_K_M	30.2 GB	2	~12–15 t/s	~30% (dense, 2-way)
Qwen3-32B Q8_0	34.8 GB	3	~10–14 t/s	~28% (dense, 3-way)
DS-R1-Distill-32B Q8_0	34.8 GB	3	~10–14 t/s	~28% (dense, 3-way)
Qwen3-30B-A3B Q8_0	32.5 GB	3	~90–95 t/s	~85% (MoE — but 2 cards was fine)
Gemma 3 27B Q8_0	28.7 GB	2	~15–20 t/s	~30% (dense, 2-way)
Seed-OSS-36B Q6_K	29.7 GB	2	~15–18 t/s	~30% (dense, 2-way)

Key insight for 3× 16GB: The third card's biggest value is enabling 70B models and running 32B at Q8_0 — NOT speeding up MoE models that already fit on 2 cards.

---

4. Inference Tools for 3-Way Pipeline Split

Every major framework supports 3-GPU pipeline parallelism. Tensor parallelism with 3 GPUs fails for most models (attention head counts not divisible by 3). **PP=3 is the universal 3-GPU strategy.**

4.1 Tool Compatibility Matrix

Tool	3-GPU Support	Mode	Configuration	Format
llama.cpp	✅ Full	Pipeline (layer split)	`-sm layer -ts 0.33,0.33,0.34`	GGUF
ik_llama.cpp (fork)	✅ Full	True tensor parallel	`-sm graph` (3–4× faster than layer split)	GGUF
Ollama	✅ Auto	Pipeline (auto-splits)	No config needed; detects 3 GPUs	GGUF
vLLM	✅ PP only	Pipeline parallel	`--pipeline-parallel-size 3`	HF/AWQ/GPTQ/FP8
ExLlamaV2	✅ Full	Layer split + TP	`--gpu_split 16,16,16` or `-gs auto`	EXL2/GPTQ
ExLlamaV3	✅ Full	TP + expert parallel	`--gpu_split auto`	EXL3/HF
SGLang	✅ PP only	Pipeline parallel	`--pp-size 3`	HF/AWQ/GPTQ/FP8
TensorRT-LLM	✅ Full	PP and TP	`--pp_size 3` (complex build process)	HF/FP8/INT4
KoboldCpp	✅ Full	Layer split	`--tensor_split 3,3,3 --gpulayers 999`	GGUF
TabbyAPI	✅ Full	Via ExLlamaV2/V3	`gpu_split: [16, 16, 16]` in config.yml	EXL2/EXL3

4.2 ik_llama.cpp — The 3-GPU Breakthrough

The ik_llama.cpp fork (January 2026) implements true tensor parallelism at the GGML graph level using NCCL for topology-aware GPU communication, keeping all GPUs at 100% utilization simultaneously. Community reports claim 3–4× performance gains over standard llama.cpp layer split mode. This fork is not yet merged into mainline llama.cpp but is the recommended tool for anyone running 3 consumer GPUs who needs maximum speed from GGUF models.

With ik_llama.cpp `-sm graph`, estimated 70B Q4_K_M speeds on 3× RTX 5060 Ti could reach **15–25 t/s** — a dramatic improvement over the 7–10 t/s from standard pipeline split. However, real-world 3-card consumer benchmarks with this fork are still scarce.

4.3 Tool Recommendations by Use Case

Use Case	Recommended Tool	Why
Quick testing	Ollama	Auto-detects 3 GPUs, zero config
Daily deployment (GGUF)	ik_llama.cpp server	Fastest 3-GPU performance via graph split
Daily deployment (GGUF, stable)	llama.cpp server	Mature, well-tested 3-GPU pipeline
Production serving	vLLM with PP=3	OpenAI-compatible API, continuous batching
Maximum quant quality	ExLlamaV3 via TabbyAPI	Best quant quality per bit (EXL3)
MoE-heavy workloads	SGLang	Expert parallelism for MoE models

---

5. Speed Comparison: Native 48GB vs 3×16GB

5.1 Llama 3.3 70B Q4_K_M Speed

Deployment	Token Generation	Prompt Processing	Notes
RTX A6000 (native 48GB)	**~14.6 t/s**	~467 t/s	768 GB/s bandwidth, single bus
RTX 6000 Ada (native 48GB)	**~18.4 t/s**	~547 t/s	960 GB/s bandwidth, single bus
3× RTX 5060 Ti (48GB pipeline)	**~7–10 t/s**	~200–300 t/s	Pipeline overhead, 3 hops
3× RTX 5060 Ti + ik_llama (graph)	**~15–25 t/s est.**	~400+ t/s	True TP, still emerging
2× RTX 4090 (48GB reference)	~19.1 t/s	~905 t/s	High-bandwidth reference
A100 80GB PCIe (datacenter reference)	~22.1 t/s	~727 t/s	Datacenter reference

Pipeline parallelism provides zero token-generation speed improvement from additional GPUs. Benchmark data is unambiguous: 2× RTX 4090 achieves 19.06 t/s on 70B Q4_K_M, 4× RTX 4090 achieves 18.83 t/s, 8× RTX 4090 achieves 18.76 t/s. Adding GPUs pools VRAM but does not accelerate single-request inference because layers execute sequentially.

5.2 Which Path for Which Workload

Workload Pattern	Best Path	Why
Single orchestrator, frontier tasks	Native 48GB	70B at 14–18 t/s, simple setup
Multiple concurrent agent roles	3×16GB	Split models across cards, serve in parallel
Background batch processing	Either	3×16GB cost-effective; native faster
Real-time interactive coding	Native 48GB	Speed matters for developer experience
Maximum model quality	Native 48GB	Fewer pipeline hops, fewer failure modes
Budget-constrained multi-model setup	3×16GB	~$1,300 vs $3,500–5,000; more flexible

---

6. Complete Agent Role Mapping — All 57 Roles at 48GB

6.1 Tier 1 — Simple Roles (8 roles) — No Upgrades

Every Tier 1 role was already well-served by 8–14B models at 32GB. The extra VRAM is wasted on these tasks.

Role	Model	Quant	Speed	Upgrade?
Router/triage	Qwen3-8B	Q4_K_M	~186 t/s	➡ No
Input validator	Qwen3-8B	Q4_K_M	~186 t/s	➡ No
Heartbeat/health	Qwen3-0.6B	Q8_0	~500+ t/s	➡ No
Notification/alert	Qwen3-8B	Q4_K_M	~186 t/s	➡ No
Sentiment analysis	Qwen3-8B	Q4_K_M	~186 t/s	➡ No
FAQ generation	Qwen3-14B	Q4_K_M	~124 t/s	➡ No
Translation	Qwen3-14B	Q4_K_M	~124 t/s	➡ No
Calendar/scheduling	Qwen3-8B	Q4_K_M	~186 t/s	➡ No

6.2 Tier 2 — Moderate Roles (25 roles) — 6 Upgrades

Role	48GB Model	Quant	Upgrade?	Why
Research/web search	**Nemotron-Super-49B**	Q4_K_M	⬆ YES	Broader knowledge, better synthesis, 32–64K ctx
Content writing	**Nemotron-Super-49B**	Q4_K_M	⬆ YES	Noticeably better prose at 49B
Editing	**Qwen3-32B**	Q8_0	⬆ YES	Near-lossless catches subtle errors
Email drafting	Qwen3-32B	Q4_K_M	➡ No	32B Q4 already excellent
Document summarization	**Qwen3-32B**	Q8_0	⬆ YES	Q8_0 + 96K context = longer docs
Meeting notes	Qwen3-30B-A3B	Q8_0	➡ Marginal	Q8_0 quality improvement, same model
Social media	Qwen3-14B	Q4_K_M	➡ No	Short-form, 14B sufficient
News aggregation	Qwen3-14B	Q4_K_M	➡ No	Classification + summary
Shopping/price comparison	Qwen3-8B	Q4_K_M	➡ No	Structured extraction
Memory/knowledge mgmt	Qwen3-30B-A3B	Q8_0	➡ Marginal	Q8_0 + extended context
RAG/retrieval	**Qwen3-30B-A3B**	Q8_0	⬆ YES	160K+ context at Q8_0 = massive RAG windows
Data analysis	**Nemotron-Super-49B**	Q4_K_M	⬆ YES	Complex data patterns need 49B+ reasoning
Website scraping	Qwen3-8B	Q4_K_M	➡ No	Parsing, not reasoning
Image description	Gemma 3 27B	Q8_0	➡ Marginal	Q8_0 was available at 32GB; now with full 128K ctx
Customer support	Qwen3-32B	Q4_K_M	➡ No	32B sufficient
Lead scoring	Qwen3-14B	Q4_K_M	➡ No	Classification task

Role	48GB Model	Quant	Upgrade?	Why
Project summarization	Qwen3-32B	Q8_0	➡ Marginal	Quality bump
Transaction/approval	Qwen3-8B	Q4_K_M	➡ No	Rule-heavy
Home automation	Qwen3-8B	Q4_K_M	➡ No	Command parsing
Fitness tracking	Qwen3-8B	Q4_K_M	➡ No	Data logging
Recipe/cooking	Qwen3-14B	Q4_K_M	➡ No	Knowledge retrieval
Personal finance	Qwen3-14B	Q4_K_M	➡ No	14B handles well
SEO optimization	Qwen3-14B	Q4_K_M	➡ No	Content analysis
Landing page generation	Qwen3-32B	Q8_0	➡ Marginal	Q8_0 quality bump
Travel planning	Qwen3-14B	Q4_K_M	➡ No	14B sufficient

6.3 Tier 3 — Complex Roles (11 roles) — 8 Upgrades

This tier sees the largest number of role upgrades. Llama 3.3 70B's HumanEval 88.4% is a substantial jump over any 32B model's ~75–80% for coding roles.

Role	48GB Model	Quant	Speed (native)	Speed (3×16GB)	Upgrade?
Code generation	**Llama 3.3 70B**	Q4_K_M	~16 t/s	~8 t/s	⬆ YES — HumanEval 88.4% vs ~78%
Code review	**Llama 3.3 70B**	Q4_K_M	~16 t/s	~8 t/s	⬆ YES — catches more subtle bugs
QA/test writing	**Llama 3.3 70B**	Q4_K_M	~16 t/s	~8 t/s	⬆ YES — better edge-case generation
Task planning	**Nemotron-Super-49B**	Q4_K_M	~20 t/s	~12 t/s	⬆ YES — reasoning toggle improves planning
Fact-checking	**Llama 3.3 70B**	Q4_K_M	~16 t/s	~8 t/s	⬆ YES — broader knowledge, better calibration
Critic/review	**Nemotron-Super-49B**	Q4_K_M	~20 t/s	~12 t/s	⬆ YES — nuanced evaluation at 49B
Market research	**Nemotron-Super-49B**	Q4_K_M	~20 t/s	~12 t/s	⬆ YES — synthesis across more data
Synthesizer/aggregator	**Qwen3-30B-A3B**	Q8_0	~34 t/s	~101 t/s (2 cards)	⬆ YES — Q8_0 + 160K ctx ideal for synthesis
Curriculum design	Qwen3-32B	Q8_0	~25 t/s	~12 t/s	➡ Marginal
Prototype generation	**Llama 3.3 70B**	Q4_K_M	~16 t/s	~8 t/s	⬆ YES — full-stack code gen benefits from 70B
DevOps	Qwen3-32B	Q8_0	~25 t/s	~12 t/s	➡ Marginal

Note on synthesizer/aggregator: Qwen3-30B-A3B at Q8_0 is preferred over a 70B model here. Its 160K+ context window at near-lossless quality makes it ideal for ingesting large volumes of text, and at ~101 t/s on 2 cards it's dramatically faster than any 70B model. Speed and context matter more than raw intelligence for aggregation tasks.

6.4 Tier 4 — Reasoning Roles (3 roles) — ALL Transformative Upgrades

The most dramatic improvement at 48GB. DeepSeek-R1-Distill-70B's AIME 2024 score of 70.0% represents a quantum leap over the 32B distill's ~50%.

Role	48GB Model	Quant	Speed (native)	Speed (3×16GB)	Upgrade?
Math/logic reasoning	**R1-Distill-70B**	Q4_K_M	~16 t/s	~8 t/s	⬆⬆ TRANSFORMATIVE — AIME 70% vs ~50%
STEM analysis	**R1-Distill-70B**	Q4_K_M	~16 t/s	~8 t/s	⬆⬆ TRANSFORMATIVE — GPQA Diamond 65.2%
Algorithm exploration	**R1-Distill-70B**	Q4_K_M	~16 t/s	~8 t/s	⬆⬆ TRANSFORMATIVE — CodeForces 1633

Tool-calling workaround: R1-Distill-70B lacks native tool calling. For roles requiring function execution, pair it with a tool-calling router: R1-Distill-70B generates the reasoning chain → Qwen3-8B or Llama 3.3 70B executes required function calls.

6.5 Tier 5 — Frontier Roles (10 roles) — 8 Upgrades

Role	48GB Model	Quant	Speed (native)	Speed (3×16GB)	Upgrade?
Orchestrator/manager	**Llama 3.3 70B**	Q4_K_M	~16 t/s	~8 t/s	⬆ YES — BFCL 77.3% critical for delegation
Software architect	**Llama 3.3 70B**	Q4_K_M	~16 t/s	~8 t/s	⬆ YES — broader system design knowledge
Complex debugger	**Llama 3.3 70B**	Q4_K_M	~16 t/s	~8 t/s	⬆ YES — HumanEval 88.4%, multi-file reasoning
Legal document review	**Nemotron-Super-49B**	Q4_K_M	~20 t/s	~12 t/s	⬆ YES — reasoning toggle + long context
Medical/health analysis	**Nemotron-Super-49B**	Q4_K_M	~20 t/s	~12 t/s	⬆ YES — broader medical knowledge
Financial analysis	**Nemotron-Super-49B**	Q4_K_M	~20 t/s	~12 t/s	⬆ YES — numerical reasoning improved
Security analyst	**Llama 3.3 70B**	Q4_K_M	~16 t/s	~8 t/s	⬆ YES — broader threat knowledge
SRE/incident response	**Llama 3.3 70B**	Q4_K_M	~16 t/s	~8 t/s	⬆ YES — tool calling + system knowledge

Role	48GB Model	Quant	Speed (native)	Speed (3×16GB)	Upgrade?
Book writing	Qwen3-30B-A3B	Q8_0	~34 t/s	~101 t/s	➡ No — 30B-A3B Q8_0 with 160K ctx optimal for long-form
Compliance/regulatory	Qwen3-32B	Q8_0	~25 t/s	~12 t/s	➡ Marginal

The orchestrator role is the most consequential upgrade. Llama 3.3 70B's BFCL 77.3% tool-calling score and IFEval 92.1% instruction-following make it substantially more reliable at meta-cognitive coordination than any 32B model. A 32B orchestrator occasionally misroutes tasks or misparses tool outputs; a 70B orchestrator does so measurably less often.

6.6 Upgrade Summary — All 57 Roles

Tier	Total Roles	Upgraded at 48GB	Unchanged	Key Driver
Tier 1 (simple)	8	0	8	8–14B already sufficient
Tier 2 (moderate)	25	6	19	Context windows + prose quality
Tier 3 (complex)	11	8	3	Code gen (HumanEval 88%) + tool calling
Tier 4 (reasoning)	3	3	0	R1-Distill-70B reasoning leap
Tier 5 (frontier)	10	8	2	70B judgment + knowledge breadth
Total	**57**	**25**	**32**	

6.7 Model Usage Across All Roles

Model	Roles Served	Tier Coverage	Why It's Used
Llama 3.3 70B Q4_K_M	10 roles	T3–T5	Highest tool-calling (BFCL 77.3%) + coding (HumanEval 88.4%). Orchestrator, coding, debugging, security, SRE, fact-checking
DS-R1-Distill-70B Q4_K_M	3 roles	T4	Reasoning specialist. AIME 70%, MATH-500 94.5%. Math, STEM, algorithms
Nemotron-Super-49B Q4_K_M	8 roles	T2–T5	Sweet spot: 70B-class quality, 32–64K context, tool calling + reasoning toggle. Research, planning, legal, medical, financial

Model	Roles Served	Tier Coverage	Why It's Used
Qwen3-30B-A3B Q8_0 (MoE)	4 roles	T2–T5	Speed king at 101 t/s (2 cards). 160K+ ctx at near-lossless. Synthesis, RAG, book writing, meetings
Qwen3-32B Q8_0	5 roles	T2–T3	Near-lossless general quality. Editing, document summary, curriculum, DevOps, landing pages
Qwen3-Coder-30B Q8_0 (MoE)	Available as alt	T3	Alternative to Llama 70B for coding when MoE speed preferred over 70B quality
Qwen3-14B Q4_K_M	12 roles	T1–T2	Mid-range single GPU. Email, social media, customer support, finance, SEO, travel
Qwen3-8B Q4_K_M	8 roles	T1–T2	Fast single GPU. Router, validator, notifications, sentiment, scheduling, scraping
Qwen3-0.6B Q8_0	1 role	T1	Heartbeat/health monitor. Ultra-lightweight

7. Deployment Architectures at 48GB

7.1 Native 48GB Card — Architectures

Architecture A — Single 70B Model: Load Llama 3.3 70B Q4_K_M (42.5 GB) with 5.5 GB for KV cache. Swap to R1-Distill-70B for reasoning tasks. Swap time ~10–20 seconds for 70B models.

Architecture B — 49B + Sidecar: Load Nemotron-Super-49B Q4_K_M (30.2 GB) + Qwen3-8B Q4_K_M (5.5 GB) simultaneously. Leaves ~12 GB for KV cache. Router on 8B model, workhorse on 49B. No swapping needed for most tasks.

Architecture C — 32B at Q8_0 + Generous Context: Load Qwen3-32B Q8_0 (34.8 GB) with 13.2 GB for KV cache (48–96K context). Near-lossless quality with massive context for document-heavy workflows.

7.2 3× RTX 5060 Ti — Architectures

Architecture A — 70B Across All 3 Cards: Llama 3.3 70B Q4_K_M split across 3 GPUs (~14.2 GB per card). Speed: ~7–10 t/s (pipeline) or ~15–25 t/s (ik_llama.cpp graph). Best for quality-critical background tasks.

Architecture B — 32B on 2 Cards + Small Model on 3rd: Qwen3-32B Q8_0 (34.8 GB) split across GPU 0 + GPU 1, Qwen3-8B Q4_K_M on GPU 2 for routing/triage. Speed: ~10–14 t/s for 32B tasks, ~186 t/s for routing. Best for multi-agent pipelines.

Architecture C — MoE Workhorse on 2 Cards + 14B on 3rd: Qwen3-30B-A3B Q8_0 (32.5 GB) split across GPU 0 + GPU 1 at ~101 t/s, Qwen3-14B Q4_K_M (9 GB) on GPU 2. Simultaneous fast MoE + mid-range

general model. Best for high-throughput mixed workloads.

Architecture D — Dual Independent + Small Router: GPU 0: Qwen3-14B Q6_K (~11 GB), GPU 1: Gemma 3 27B QAT Q4_0 (~15.6 GB), GPU 2: Qwen3-8B Q4_K_M (~5.5 GB) for routing. Three simultaneous models, zero swap latency. Best for diverse multi-agent pipelines.

7.3 Recommended Architecture by Workload

Workload	Native 48GB	3× RTX 5060 Ti
Heavy coding sessions	Arch A: Llama 70B all day	Arch A: 70B on 3 cards (slower)
Multi-agent pipeline	Arch B: 49B + 8B sidecar	Arch C: MoE on 2 + 14B on 3rd
Document-heavy analysis	Arch C: 32B Q8_0 with huge ctx	Arch B: 32B Q8_0 on 2 + router on 3rd
Deep reasoning tasks	Swap to R1-Distill-70B	Arch A: R1-Distill-70B on 3 cards
Mixed diverse roles	Arch B: 49B + 8B	Arch D: 3 independent models

---

8. Models That Do NOT Fit at 48GB

No CPU offloading. No partial fits. If it exceeds 48GB in weights, it's out.

Model	Params	Architecture	Smallest Viable Quant	Weight Size	Verdict
DeepSeek-R1	671B	MoE	Q2_K	~250 GB	❌ Datacenter only (8× H100)
DeepSeek-V3 / V3.2	685B	MoE	Q2_K	~260 GB	❌ Datacenter only
Qwen3-235B-A22B	235B	MoE	Q2_K	~85.7 GB	❌ Needs 2× 48GB minimum
GPT-oss-120B	117B	MoE	MXFP4	~59 GB	❌ 11 GB over limit
Llama 4 Scout	109B	MoE	Q4_K_M	~65 GB	❌ 17 GB over limit
Llama 4 Scout	109B	MoE	IQ2_XXS	~30–37 GB	❌ Fits VRAM but quality destroyed at 2-bit — unusable
Llama 4 Maverick	400B	MoE	IQ1_S	~122 GB	❌ Not remotely close
Command R+	104B	Dense	Q3_K_M	~51 GB	❌ 3 GB over limit
Mixtral 8x22B	141B	MoE	Q4_K_M	~80 GB	❌ 32 GB over limit
Llama 3.3 70B	70B	Dense	Q5_K_M	49.95 GB	❌ 2 GB over limit
Qwen2.5-72B	72.7B	Dense	Q4_K_M	47.4 GB	❌ Fits weights but 0.6 GB KV = unusable
Nemotron-Super-49B	49B	Dense	Q8_0	53 GB	❌ 5 GB over limit

9. Conclusion & Recommended OpenClaw Model Stack at 48GB

9.1 Primary Recommendation

Priority	Model	Roles Covered	Key Advantage
Frontier workhorse	Llama 3.3 70B Q4_K_M	10 roles (T3–T5): orchestrator, coding, debugging, security, SRE, fact-checking	BFCL 77.3% + HumanEval 88.4%

Priority	Model	Roles Covered	Key Advantage
Deep reasoning	DS-R1-Distill-70B Q4_K_M	3 roles (T4): math, STEM, algorithms	AIME 70%, MATH-500 94.5%
Analytical sweet spot	Nemotron-Super-49B Q4_K_M	8 roles (T2–T5): research, planning, legal, medical, financial	70B-class quality + 32–64K ctx + tool calling
Fast MoE general	Qwen3-30B-A3B Q8_0	4 roles (T2–T5): synthesis, RAG, book writing	101 t/s + 160K+ ctx at near-lossless
Quality general	Qwen3-32B Q8_0	5 roles (T2–T3): editing, documents, curriculum	Near-lossless quality at 48GB
Mid-range	Qwen3-14B Q4_K_M	12 roles (T1–T2): email, support, social	Fast single-GPU
Fast lightweight	Qwen3-8B Q4_K_M	8 roles (T1): routing, validation, alerts	186 t/s

9.2 Core Insight: 48GB Is Where Local Matches Cloud

The 70B class brings three capabilities absent at 32GB:

- **Tool-calling reliability above 77% BFCL** — critical for the orchestrator delegating to all other agents
- **HumanEval code generation above 88%** — lifting all coding agent roles substantially
- **Mathematical reasoning at AIME 70%** — making Tier 4 roles genuinely capable rather than aspirational

48GB is the first tier where local inference stops being a compromise and starts being a genuine alternative to cloud APIs for production agent workloads.

9.3 Minimum Viable Stack

For the simplest effective OpenClaw deployment at 48GB:

1. **Nemotron-Super-49B Q4_K_M** (30.2 GB) — covers orchestrator, planning, research, coding, reasoning-toggle tasks. Fits with 17.8 GB for 32–64K context. Handles majority of T2–T5 roles.
2. **Qwen3-8B Q4_K_M** — fast router for all Tier 1 roles
3. Swap to **Llama 3.3 70B Q4_K_M** for heavy coding sessions or complex orchestration requiring maximum tool-calling reliability
4. Swap to **R1-Distill-70B Q4_K_M** for deep reasoning tasks

This rotation covers all 57 roles. Nemotron-Super-49B as the default workhorse handles 80%+ of tasks with its combination of quality, tool calling, reasoning toggle, and manageable VRAM footprint.

9.4 Honest Assessment: What 48GB Does NOT Solve

- Cloud APIs (Claude, GPT-4.1, Gemini 2.5 Pro, o3) still outperform on the hardest multi-step reasoning
- Full DeepSeek-R1 (671B) and DeepSeek-V3 quality remains datacenter-only
- Legal, medical, and compliance roles still benefit from cloud verification on critical decisions
- Context windows at 70B Q4_K_M (8–16K) are tight — use Nemotron-Super-49B Q4_K_M (32–64K) when context matters

25 of 57 roles upgrade at 48GB. All Tier 4 reasoning roles transform. The remaining 32 roles were already well-served at 32GB. The upgrades concentrate in Tiers 3–5 where task complexity exceeds what 32B models handle reliably.